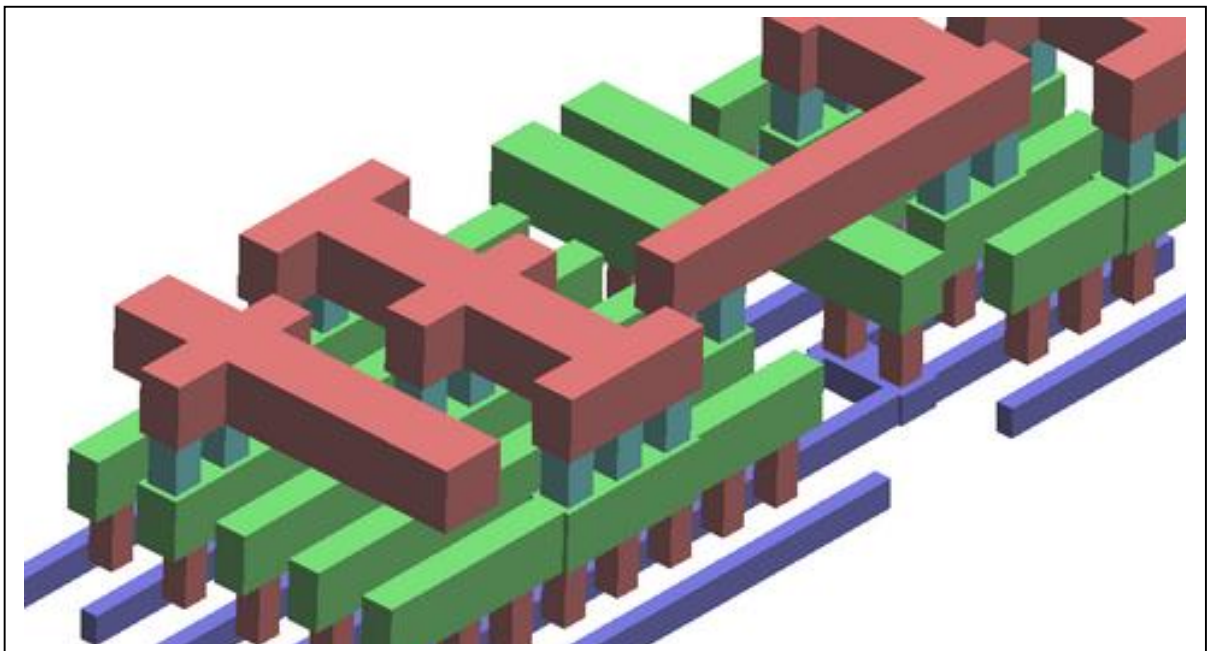


Eng. Ali Eltemsah Diploma

PnR Lab

Fully Detailed Report



By: Khaled Ahmed Hamed

This is our directory:



Open DFT Folder

Prepare different constraints Files:

- Copy ALU_TOP.sdc to ALU_TOP_func.sdc and add the following commands at the end:
 - a. set_case_analysis 0 test_mode
 - b. set_case_analysis 0 SE

```
set_case_analysis 0 [get_port test_mode]
set_case_analysis 0 [get_port SE]
```

- Copy ALU_TOP.sdc to ALU_TOP_scan.sdc and add the following:-
 - c. set_case_analysis 1 test_mode
 - d. set_case_analysis 1 SE

```
set_case_analysis 1 [get_port test_mode]
set_case_analysis 1 [get_port SE]
```

- Copy ALU_TOP.sdc to ALU_TOP_capture.sdc and add the following:-
 - e. set_case_analysis 1 test_mode
 - f. set_case_analysis 0 SE

```
set_case_analysis 1 [get_port test_mode]
set_case_analysis 0 [get_port SE]
```

Open PnR Folder then Import Folder then MMC.tcl (Open any file with gedit)

- Create Library sets
- Create constraints mode using create_constraint_mode for ALU_TOP_scan.sdc, ALU_TOP_func.sdc and ALU_TOP_capture.sdc
- Create RC Corner
- Create Delay Corners (Min and Max corners)
- Create analysis views for the 3 constraints modes (Setup and hold for each)
- Set analysis view for the 6 views
- **"../std_cells/ : Relational Path**

ده المكان اللي انا واقف فيه هرجع خطوة لورا بعدها اروح للستاندرد سيلز

```

create_library_set -name min_library -timing "../std_cells/libs/scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c.lib"
create_library_set -name max_library -timing "../std_cells/libs/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.lib"
create_library_set -name typ_library -timing "../std_cells/libs/scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.lib"

create_constraint_mode -name func_mode -sdc_files {../dft/ALU_TOP_func.sdc}
create_constraint_mode -name scan_mode -sdc_files {../dft/ALU_TOP_scan.sdc}
create_constraint_mode -name capture_mode -sdc_files {../dft/ALU_TOP_capture.sdc}

create_rc_corner -name RCcorner -cap_table "../std_cells/captables/tsmc13fsg.capTbl"

create_delay_corner -name min_corner -library_set min_library -rc_corner RCcorner
create_delay_corner -name max_corner -library_set max_library -rc_corner RCcorner

create_analysis_view -name setup1_analysis_view -delay_corner max_corner -constraint_mode func_mode
create_analysis_view -name hold1_analysis_view -delay_corner min_corner -constraint_mode func_mode

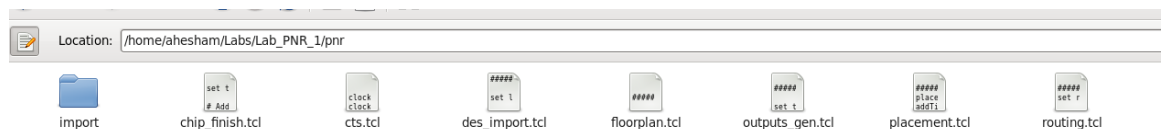
create_analysis_view -name setup2_analysis_view -delay_corner max_corner -constraint_mode scan_mode
create_analysis_view -name hold2_analysis_view -delay_corner min_corner -constraint_mode scan_mode

create_analysis_view -name setup3_analysis_view -delay_corner max_corner -constraint_mode capture_mode
create_analysis_view -name hold3_analysis_view -delay_corner min_corner -constraint_mode capture_mode

set_analysis_view -setup {setup1_analysis_view setup2_analysis_view setup3_analysis_view} -hold {hold1_analysis_view
hold2_analysis_view hold3_analysis_view}

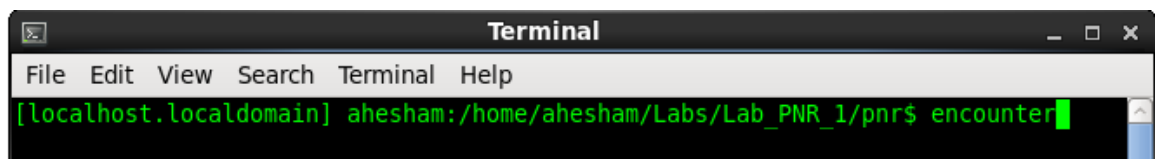
```

Open PnR folder

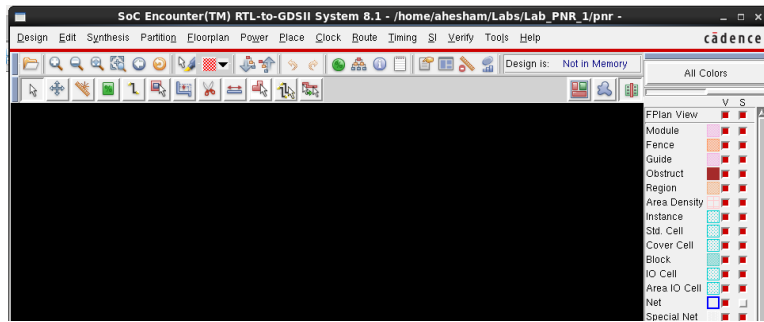


This is PnR folder at start

Open terminal and then type: encounter and Press Enter

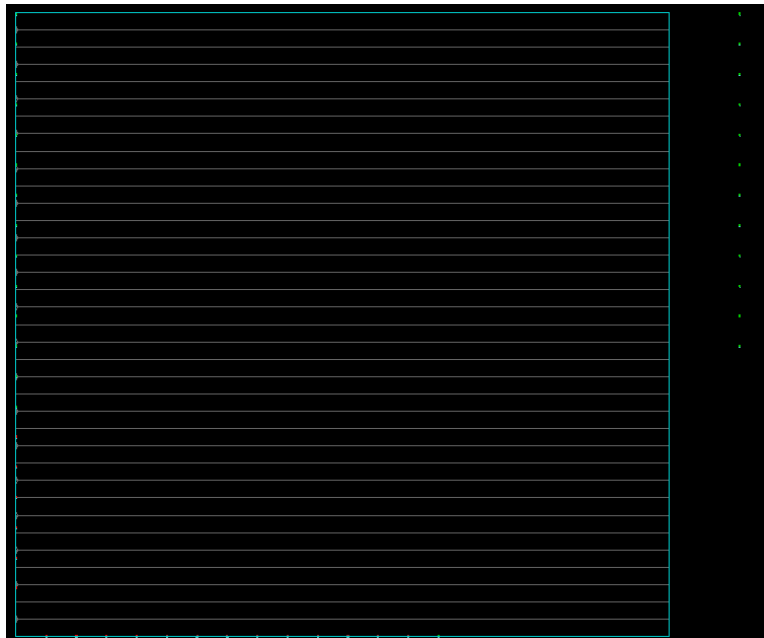


The tool will open



Type: source des_import.tcl in terminal:

```
encounter 2> source des_import.tcl
Loading Lef file /home/ahesham/Labs/Lab_PNR_1/std_cells/lef/tsmc13fsg_7lm_tech.l
ef...
Loading Lef file /home/ahesham/Labs/Lab_PNR_1/std_cells/lef/tsmc13_m_macros.lef.
..
Set DBUPerIGU to M2 pitch 820.
Initializing default via types and wire widths ...
Loading Lef file /home/ahesham/Labs/Lab_PNR_1/pnr/import/ALU.lef...
**WARN: (SOCLF-246): The 'UNITS' attribute should be set
in the first lef file (technology lef). There is an attempt to set it
in subsequent lef files which will be ignored.
**WARN: (SOCLF-46): Class CORE macro 'ALU TOP' has no SITE statement defined
.
Class CORE macros require a SITE statement. The SITE FE CORE 0 was
created and will be used for this macro, using height 120.1300 that
matches the macro SIZE height, and width 0.4100 that matches the
```



Type: source floorplan.tcl in terminal:

```
Terminal
File Edit View Search Terminal Help
Total number of always on buffers: 0
Total number of retention cells: 0
List of usable buffers: BUF10M BUF12M BUF14M BUF16M BUF18M BUF20M BUF24M
BUF26M BUF32M BUF36M BUF40M BUF44M BUF48M BUF52M BUF56M BUF60M BUF64M
BUF68M BUF72M BUF76M BUF80M BUF84M BUF88M BUF92M BUF96M BUF100M
List of unusable buffers:
Total number of usable buffers: 26
List of unusable buffers:
Total number of unusable buffers: 0
List of usable inverters: CLKINV12M CLKINV16M CLKINV20M CLKINV24M
CLKINV28M CLKINV32M CLKINV36M CLKINV40M CLKINV44M CLKINV48M CLKINV52M
CLKINV56M CLKINV60M CLKINV64M CLKINV68M CLKINV72M CLKINV76M CLKINV80M
CLKINV84M CLKINV88M CLKINV92M CLKINV96M CLKINV100M
List of unusable inverters:
Total number of unusable inverters: 0
List of identified usable delay cells: DLY1X1M DLY1X4M DLY2X1M DLY2X4M DLY3X1M
DLY3X4M DLY4X1M DLY4X4M
Total number of identified usable delay cells: 8
List of identified unusable delay cells:
Total number of identified unusable delay cells: 0
encounter 3> source floorplan.tcl
Adjusting Core to Left to: 3.2800, Core to Bottom to: 3.2800.
encounter 4>
```

```
set load_fp 0

set top_module ALU_TOP

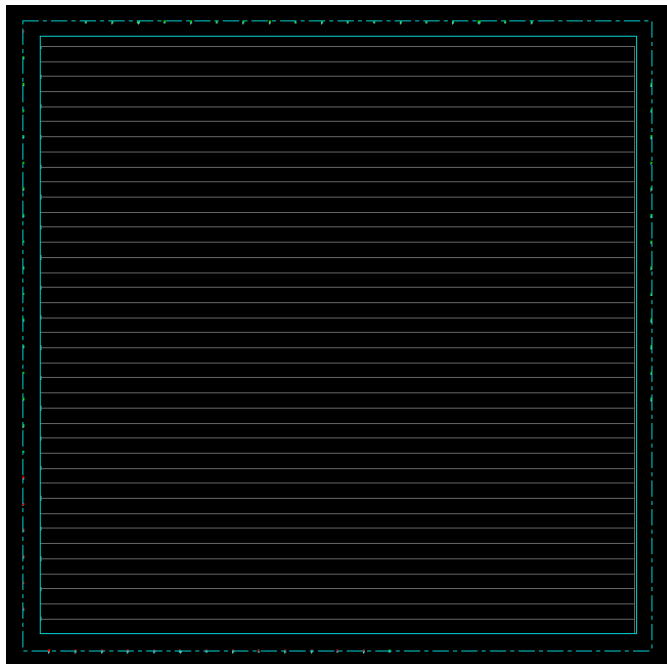
##### Define Aspect Ratio (Length/Width) of Digital Macro #####
if {$load_fp == 0} {

floorPlan -d 120.13 120.13 3.0 3.0 3.0 3.0

} else {

loadFPlan ./ $top_module.fp
```

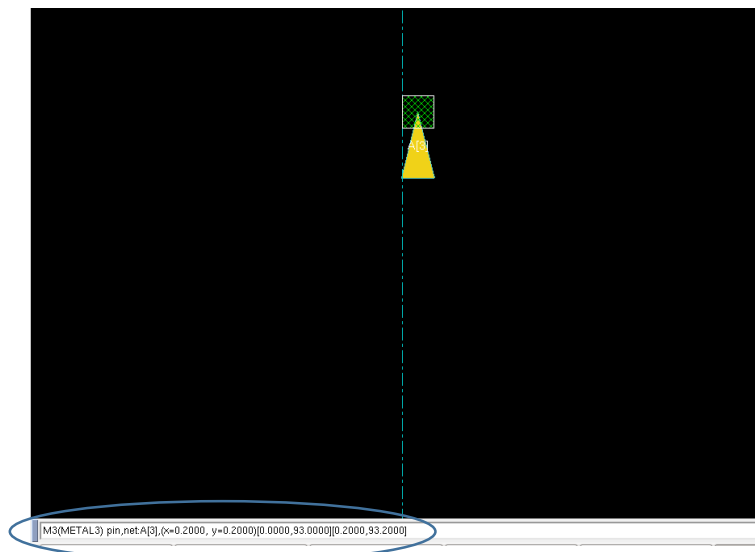
At first time only use load_fp with 0, After making a Floor Plan and if you you want to load it make it equal 1



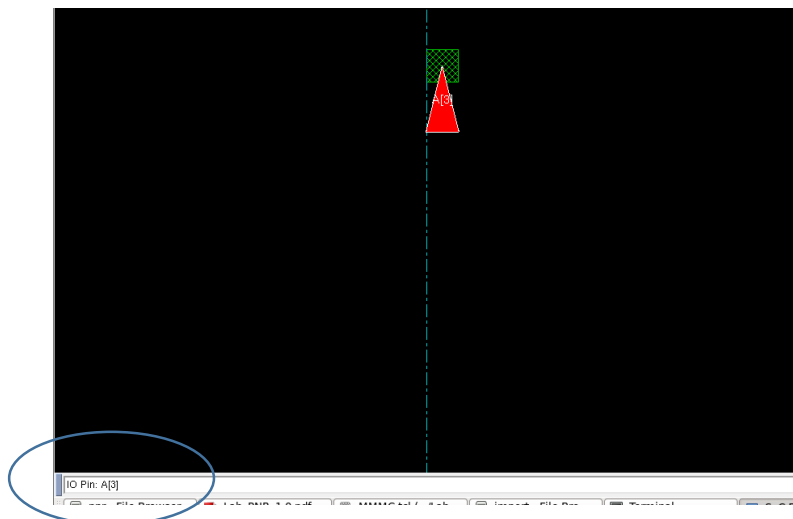
Zoom in by right clicking and selecting needed area:



Click on green square (Metal 3):



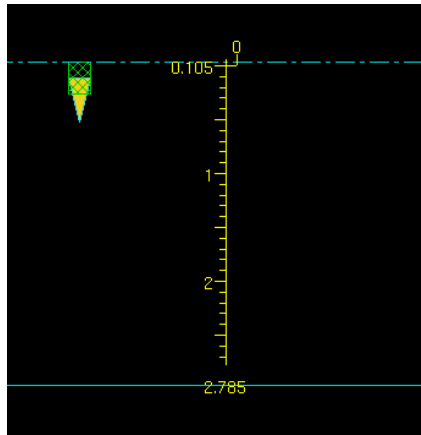
Click on pin:



To return to fit mode press **F** on keyboard

We are at select mode(a), type k on keyboard or from tool bar select ruler

Measure the distance you want

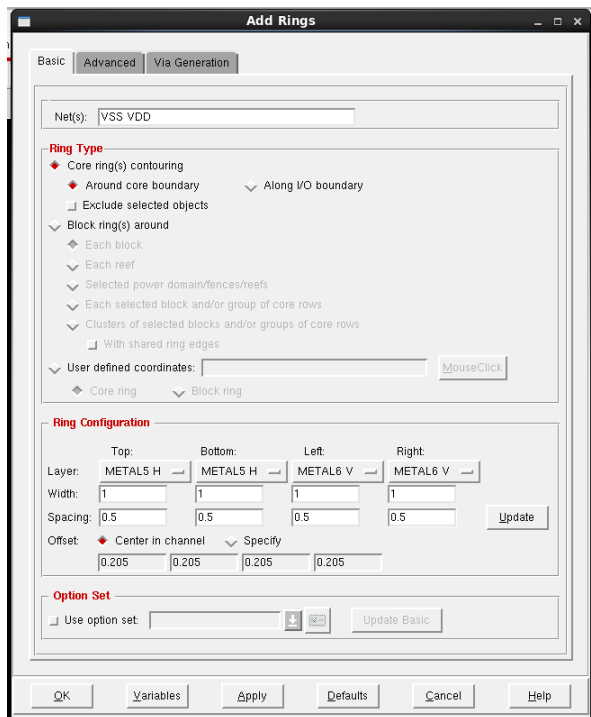


When you finish clear ruler



Power Planning:

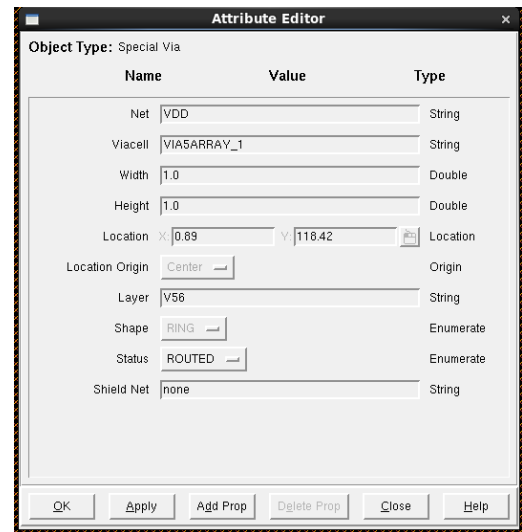
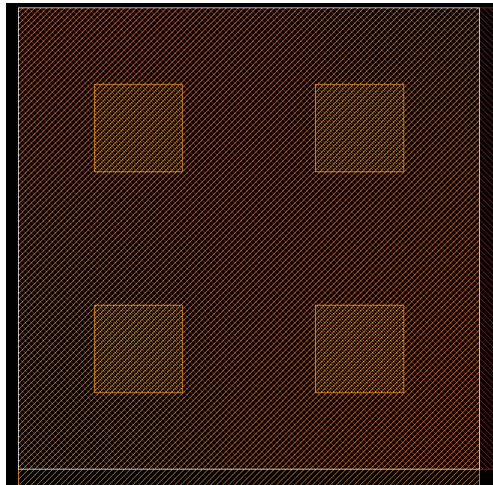
From the above bar select power then power planning then add rings



- We have 7 layers, leave the 7th layer for Analog engineer, modify top and bottom layers to be nearest horizontal from end (5th) and modify left and right layers to be nearest vertical from end (6th)
- Modify width and spacing
- Make offset center in channel



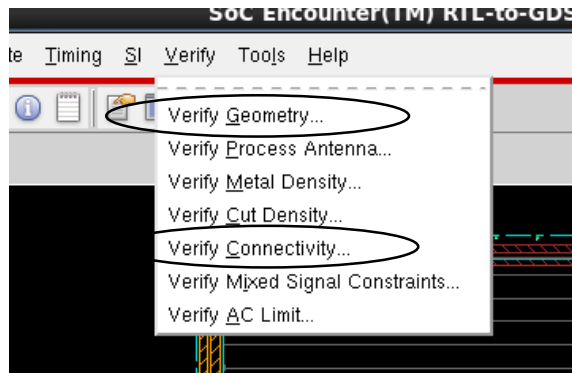
Select via then press Q



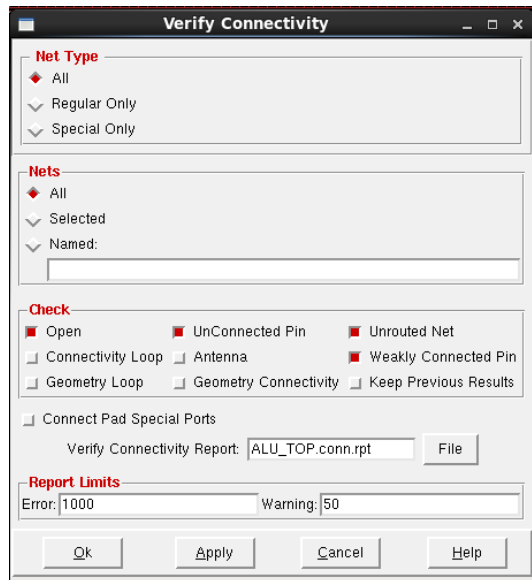
It's a via from 5 to 6, if you want to route 3 to 6 you need 3 vias (3->4, 4->5, 5->6)

ساعات التوول مش بتوصلهم كلهم صح وتيجي عند اخر واحدة وتكراش لو الديزاين كبير ساعتها هنمسحه ونعملهم مانيوال

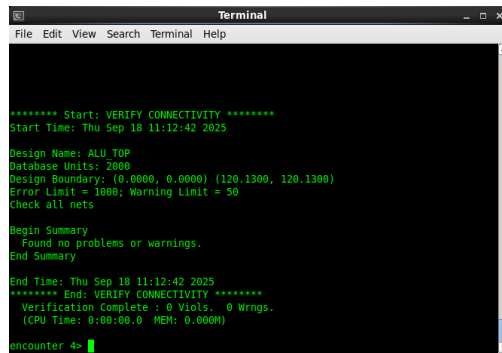
After any wire operation must check 2 things to check DRC violations: Connectivity and Geometry



First click verify connectivity:



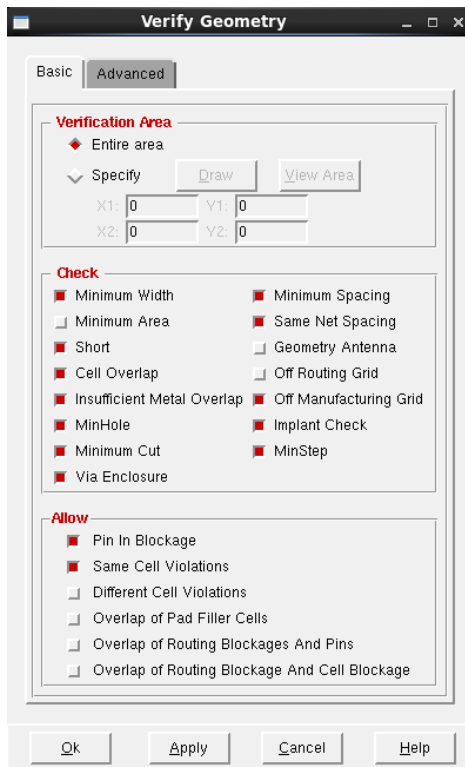
Remove check from Antenna as it checks pads and we don't have pads (Silly Violations) then open terminal



0 Violations and 0 Warnings

احنا كدة في السليم

Second click verify geometry:



Remove check from Minimum Area as it checks pads and we don't have pads (Silly Violations) then open terminal

```
Terminal
File Edit View Search Terminal Help
VERIFY GEOMETRY ..... Creating Sub-Areas
..... bin size: 8000
VERIFY GEOMETRY ..... SubArea : 1 of 1
VERIFY GEOMETRY ..... Cells : 0 Viols.
VERIFY GEOMETRY ..... SameNet : 0 Viols.
VERIFY GEOMETRY ..... Wiring : 9 Viols.
VERIFY GEOMETRY ..... Antenna : 0 Viols.
VERIFY GEOMETRY ..... Sub-Area : 1 complete 9 Viols. 0 Wrngs.
VG: elapsed time: 0.00
Begin Summary ...
Cells : 0
SameNet : 0
Wiring : 9
Antenna : 0
Short : 0
Overlap : 0
End Summary

Verification Complete : 9 Viols. 0 Wrngs.

*****End: VERIFY GEOMETRY*****
*** verify geometry (CPU: 0:00:00.1 MEM: 1.0M)
encounter 4>
```

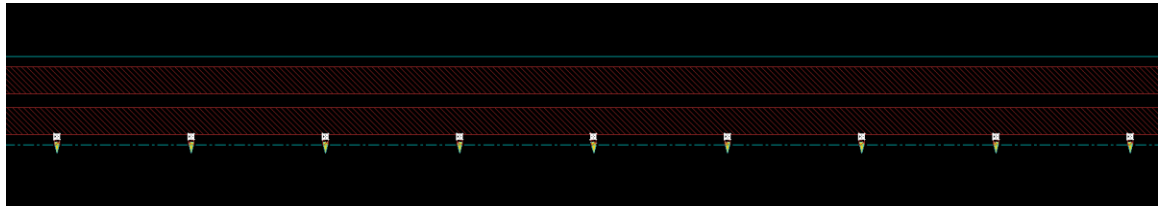
9 Violations and 0 Warnings

محتاجين نحل المشكلة دي



الإكسات دي اماكن الفيوليشنز, بيقا محتاجين نحل المشكلة في المنطقة دي بعدها نرجع ننشيك تاني جيوميتري
وكونيكتيفيتي

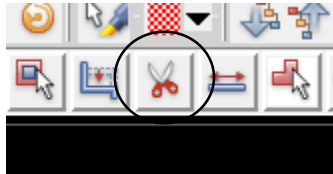
Zoom in to reach violated area



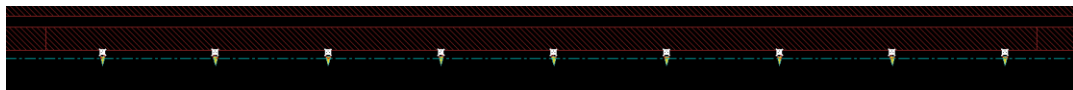
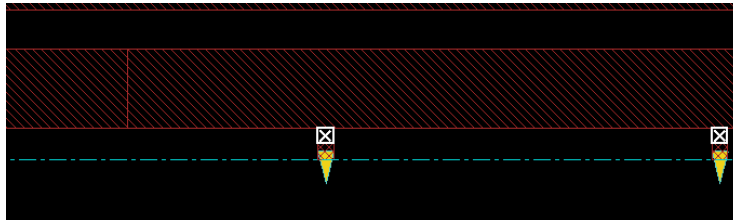
Minimum spacing violation

٢ كونتاكت علي نفس الميتال بينهم فيوليشن فمممكن نحل المشكلة دي باننا نقطعها وننزل الحثة دي علي ميتال ٣
(اقرب واحد هوريزنتال من خمسة)

Select cut wires or type x

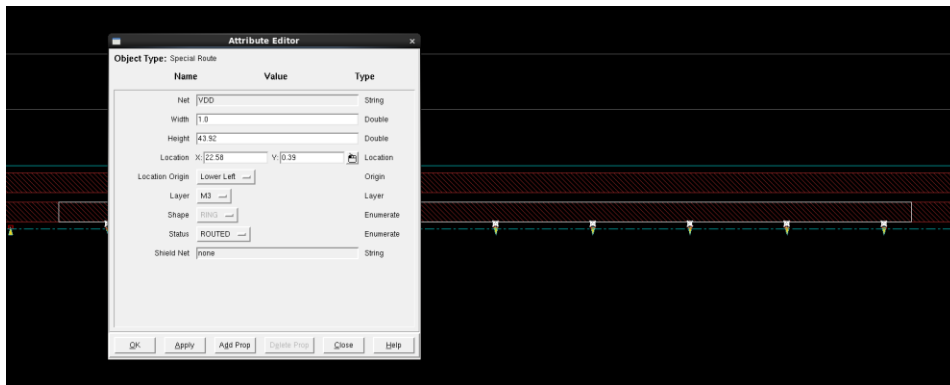


اقطع يمين وشمال زي كدة

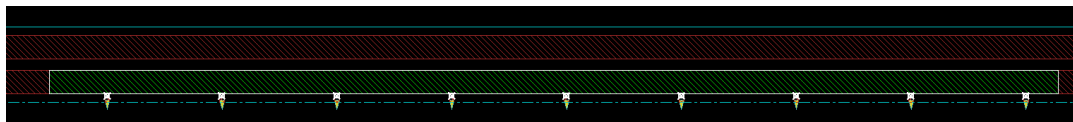


Click a

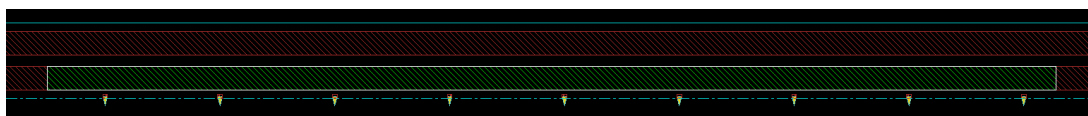
Press on cut wire then press q



Make it layer 3



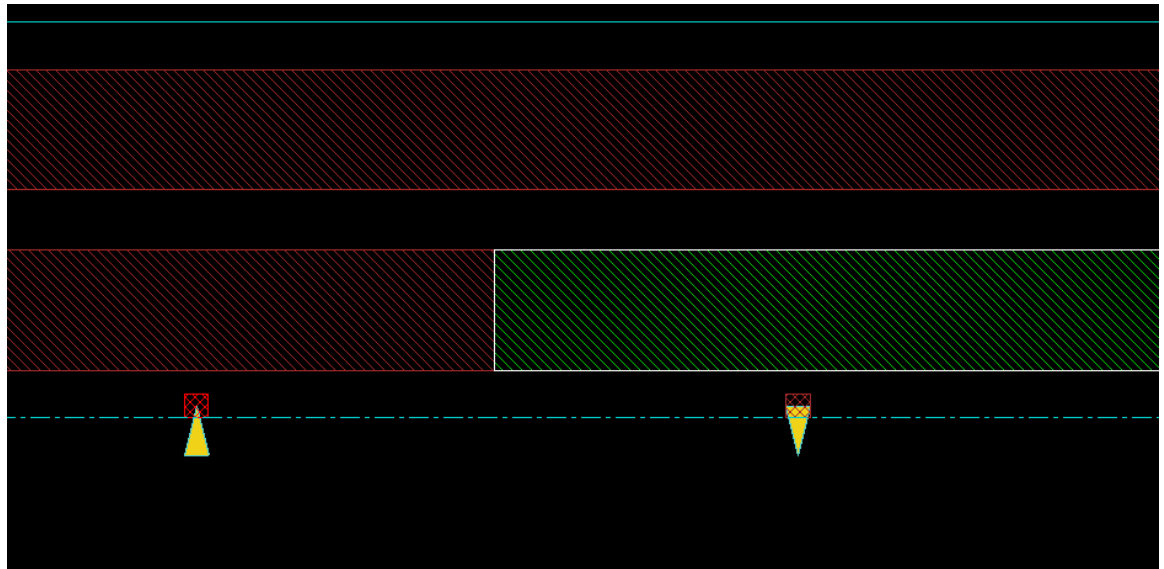
You get this, Click X



You get this

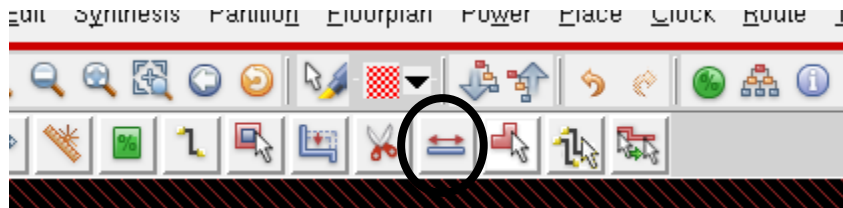
دلوقتي جزء من الواير علي ميتال ٣ و جزء علي ٥ فمحتاجين نخط فيا

Zoom in to reach this at left cut wire corner

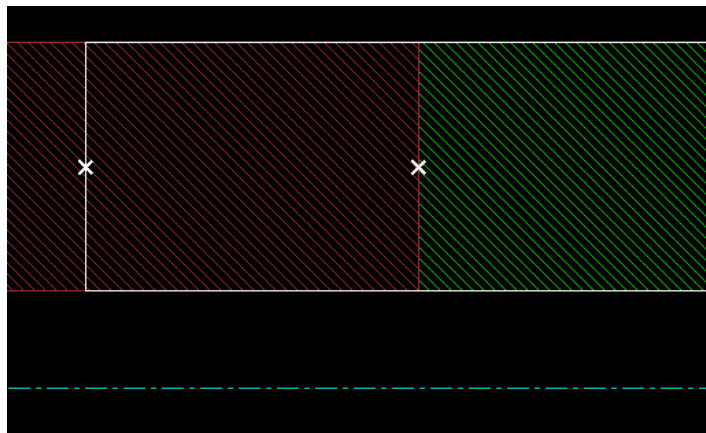


شد ميتال ٣ خليه فوق ٥ او العكس (الاحسن تشد ٣)

Click s or select stretch mode from tool bar

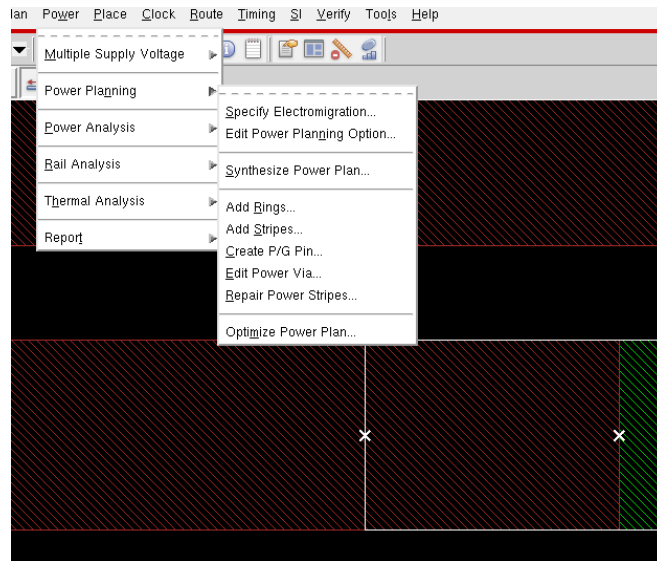


هتقف علي طرف ميتال ٣ هيظهر سهم كدة تشده علي قد مساحة الفيا اللي انت عايزها

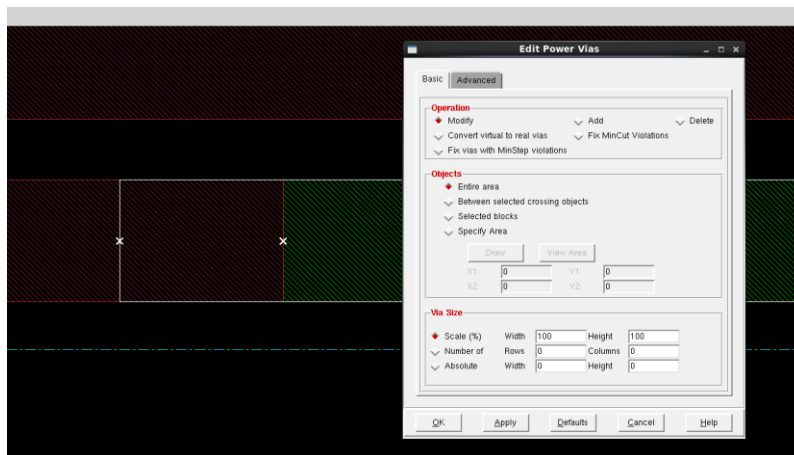


عايزين نخط الفيا بقا

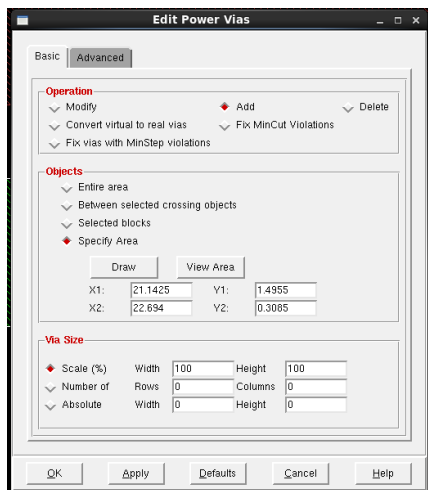
Select power planning then edit power via

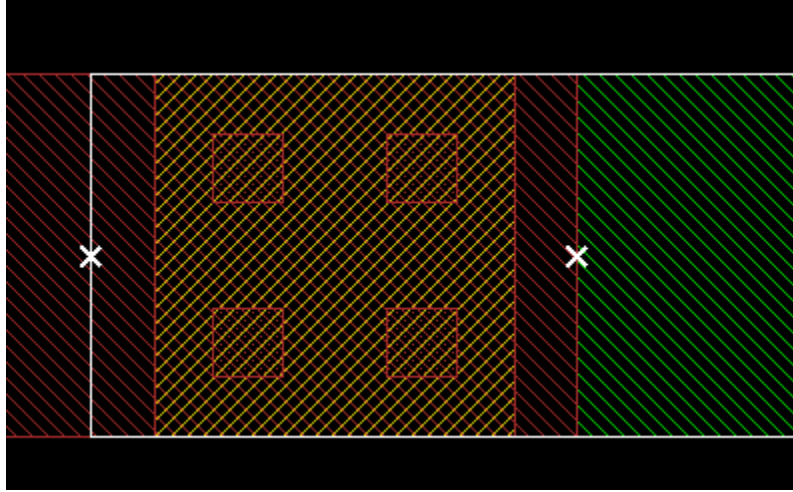


خلي الويندو علي اليمين والحتة اللي عايزين نخط فيها فيا علي الشمال زي كدة

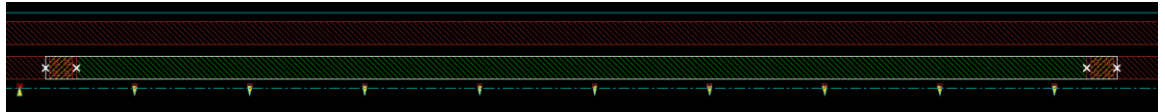


Modify check boxes, Add, specify Area and draw then select the needed area then press ok



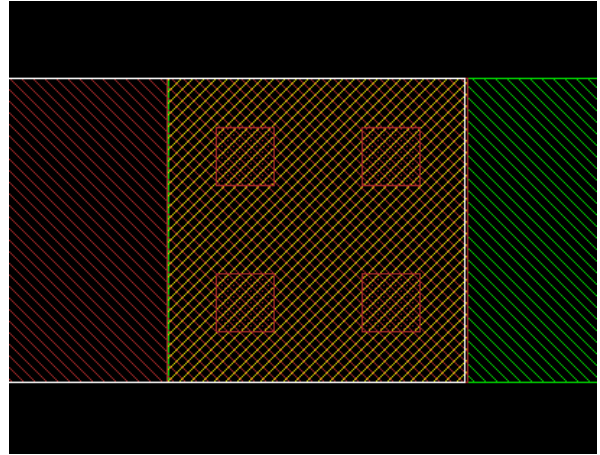
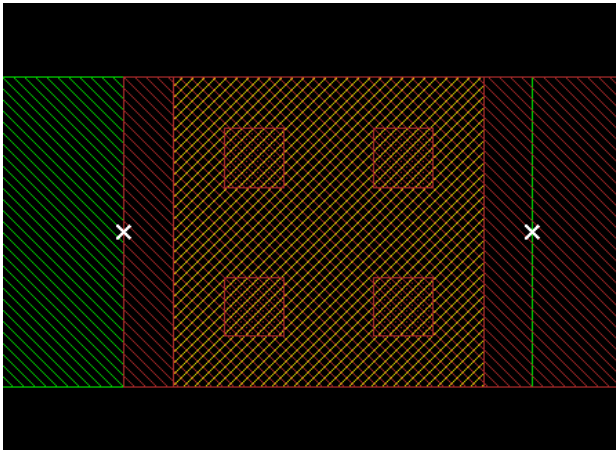


You get this, repeat it a second time in the other side



هناقي هنا الوابر مش واصل لآخر الفيا وفي اكسات يعني في فيوليشنز فمحتاجين نمده شوية بحيث يكون الوابر والفيا علي قد بعض

ادخل الستريتش مود وحرك الوابر والفيا بحيث يكونوا فيت علي بعض



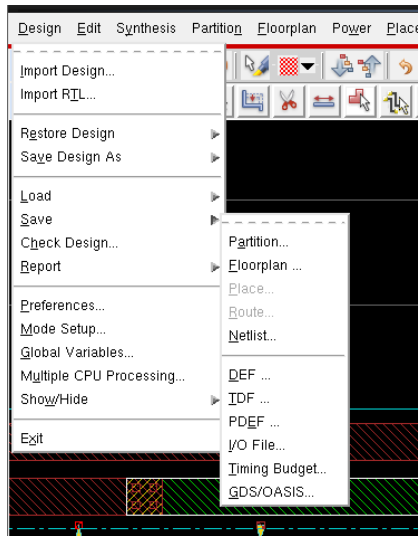
خلي بالك وانت في الستريتش مود تبوظ حاجة, خلص اللي انت عايزه وخش في السيلكت مود

تفتكر المفروض نعمل ايه دلوقتي ياعزيزي؟

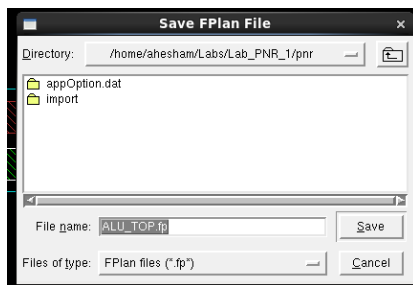
مضبوط نتشيك كونيكثيفيتي وجيومتري والمفروض يكونوا دلوقتي

Zero violations and zero warnings

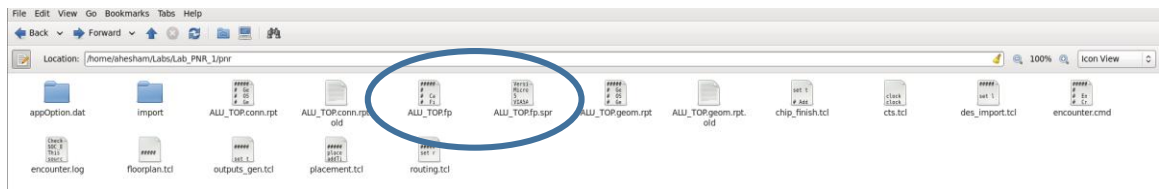
Now we want to save floor plan



Design -> Save -> Floorplan

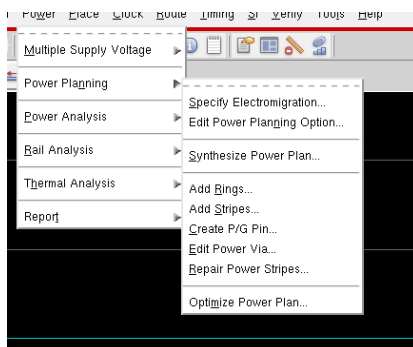


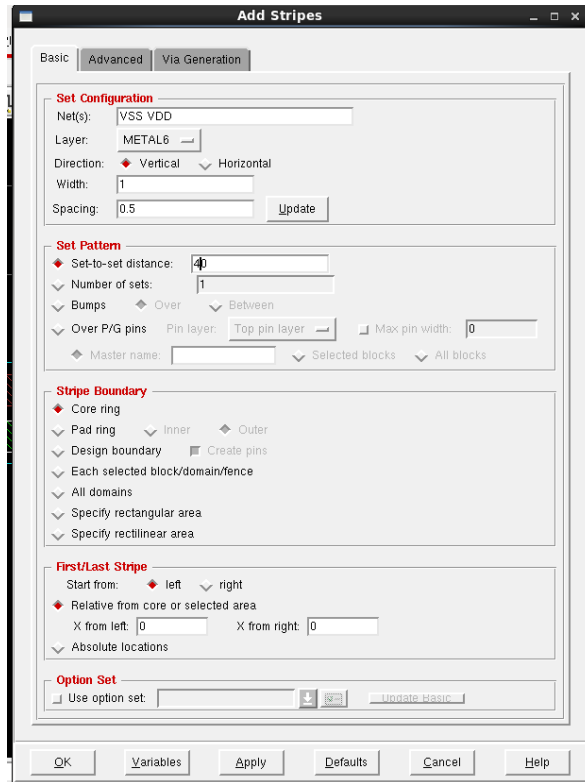
We get 2 files: ALU_TOP.fp and ALU_TOP.spr



دلوقتي عايزين نخط ستر ابيس

Power -> Power planning -> Add stripes

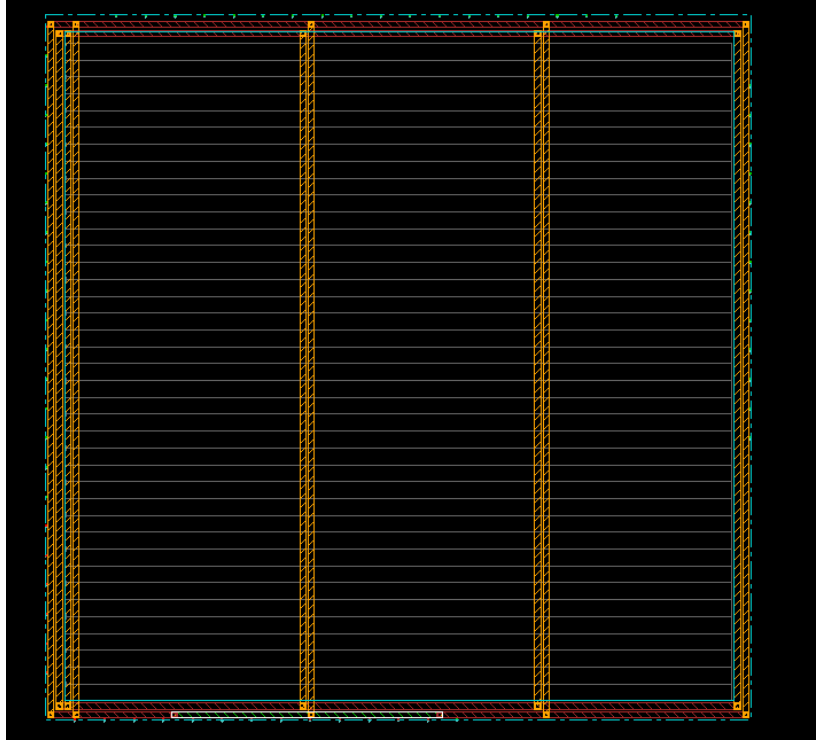




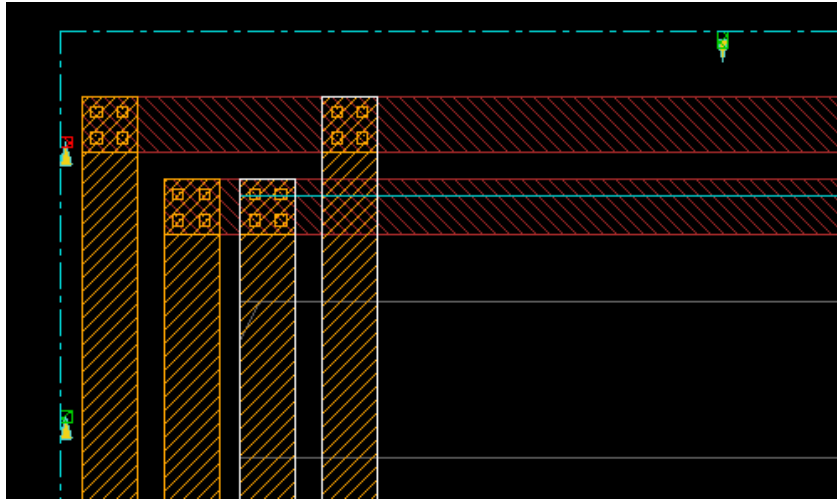
السترايبس دي بالطول فعازينها علي ميتال ٦ , زود الويدث والسبيسنج زي ماعملنا قبل كدة

Set to set distance = 40

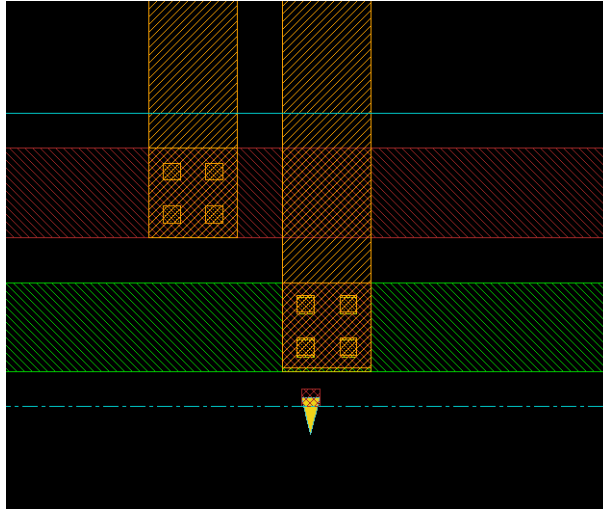
كل اتنين جمب بعض اسمهم سيت, فكدا خلي المسافة بين كل سيت والتانية ٤٠ فكدة هيقسم العرض بتاعك ١٢٠ عليهم ويوزعهم او انك تخلي عدد السيتس في الباترن ٣ او اي رقم ثاني انت عايزه



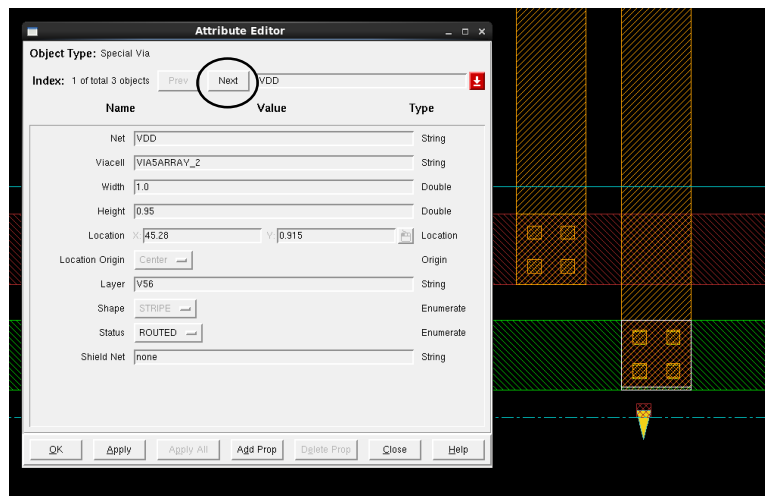
هنلاقي انه ضاف فعلا السترايبس بس في ٢ زيادة جمب الرينج لازم نشيلهم, روح عندهم وسيلكت ودوس ديليت



بعد كل خطوة زي ماتفقنا ياعزيزي لازم نتشيك كونيكتيفيتي وجيومتري
تعالا بقا هنا عند السترايب والفيا وعايزين نتأكد ان شغلنا صح

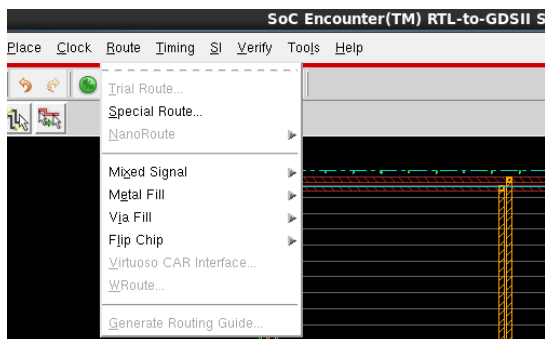


Select via the press q

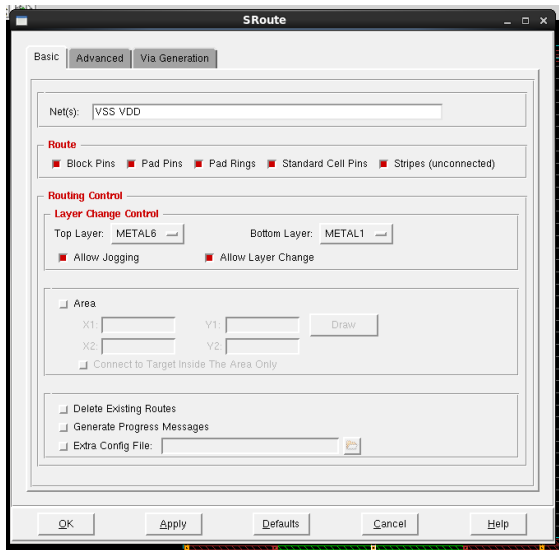


كدة عندنا فعلا ٣ فيا من ٥ ل ٦ ومن ٤ ل ٥ ومن ٣ ل ٤ فاحنا في التمام
كدة ناقصنا اخر حاجة في الفلور بلان وهي السيشال راوتنج او الخطوط اللي بالعرض

Special Routing



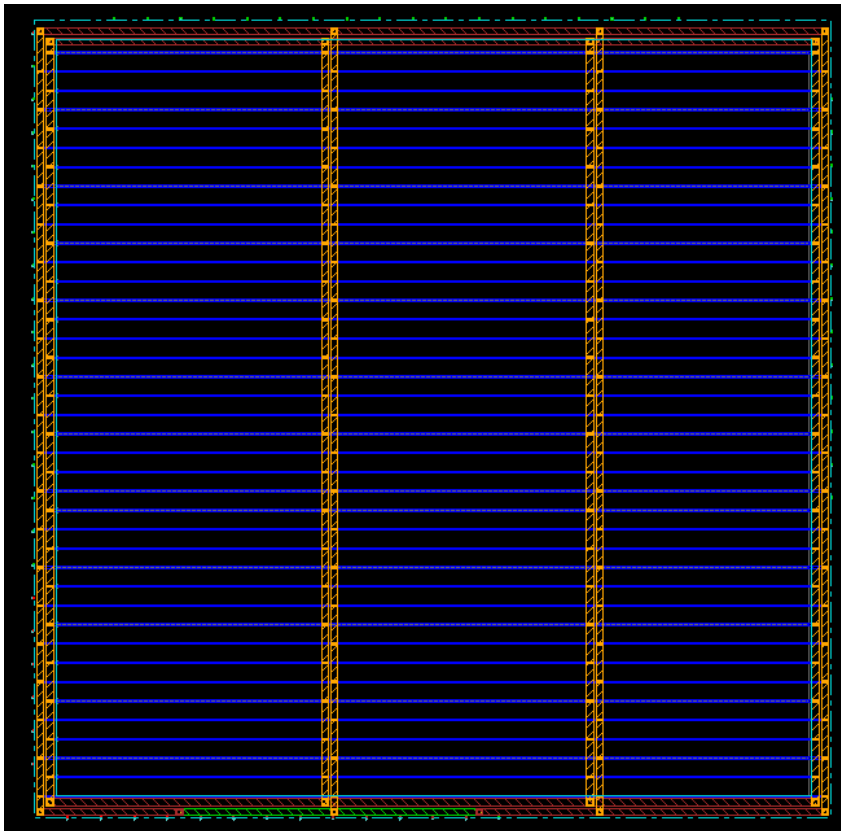
Select Route from tool bar then Special Route



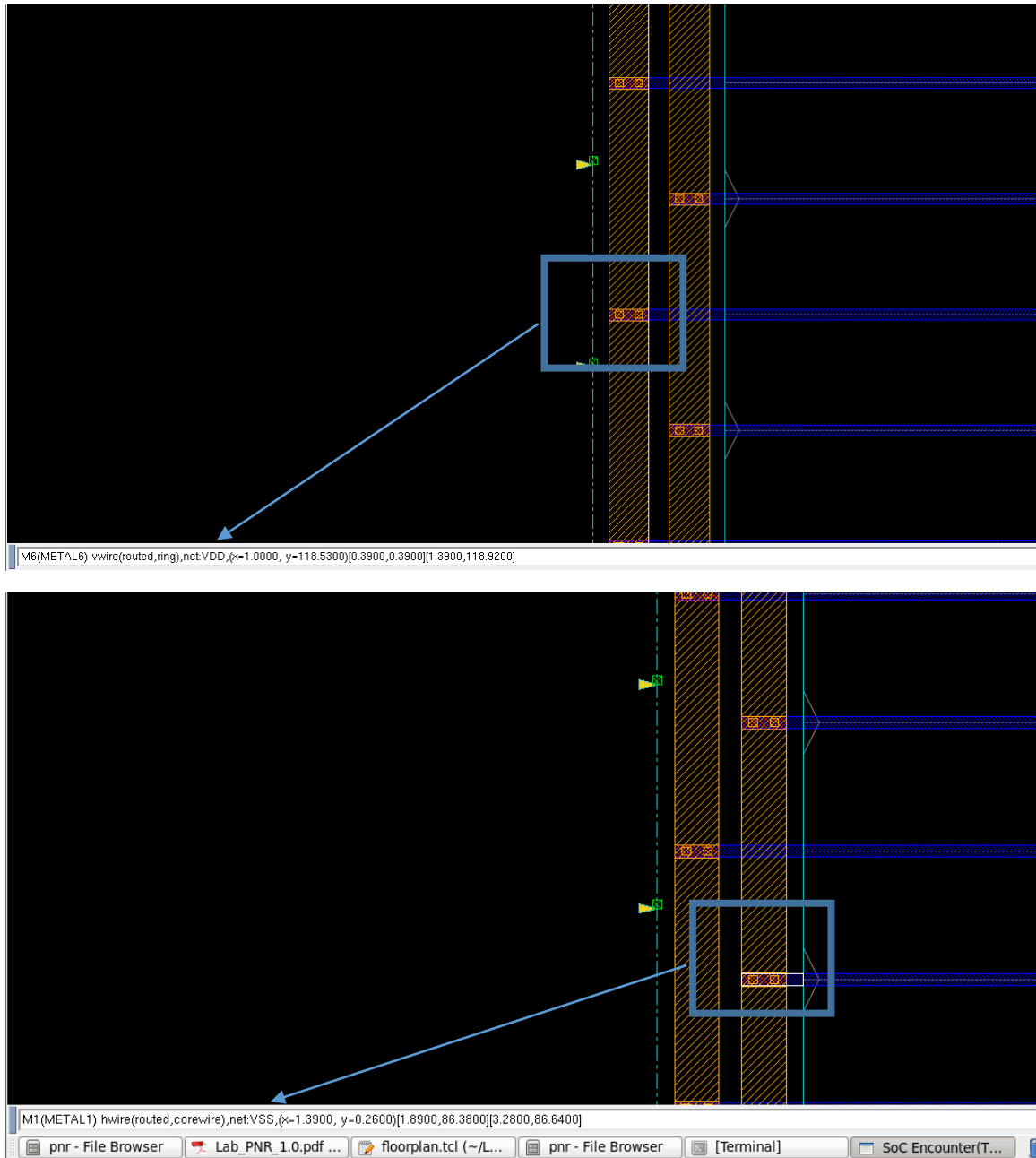
Top layer -> Metal 6

زي ماتفقنا سايبين اخر لاير خالص مش بنقرب منها

VSS & VDD: these names come from .lef library



You get this



فهيبتقا ماشية كدة واحدة واحدة

وزي ماتعودنا ياعزيزي فيريفاي جيومتري وكونيكتيفيتي بعد اي خطوة

Now save floor plan and overwrite on the old one

دلوقتني هنقل التول ونجرب نفتح ونعمل لودينج للفلور بلان ونصلي علي النبي وندعي انها تشتغل

قبل مانعمل كدة عابزين نقول كام حاجة

ALU_TOP.fp & ALU_TOP.spr

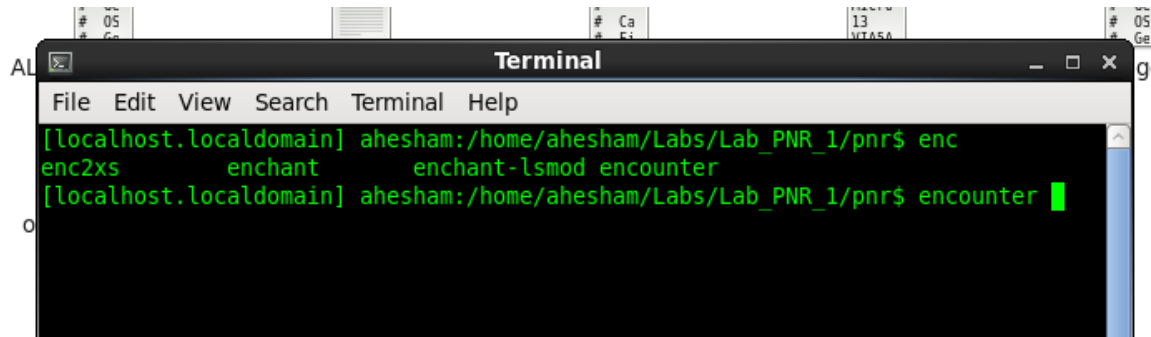
الدوت اف بي معمول جواه سورسينج للدوت اس بي ار وجواه معلومات الديزاين كلها من الرو كيريشن والكوردينتس وهكذا

والدوت اس بي ار جواه معلومات الباور ابلان اللي لسة عاملينها دلوقتى

دلوقتى جينا للحظة اننا نعمل لودينج للفلور بلان:

Open des_import.tcl and floorplan.tcl the make load_fp equal 1

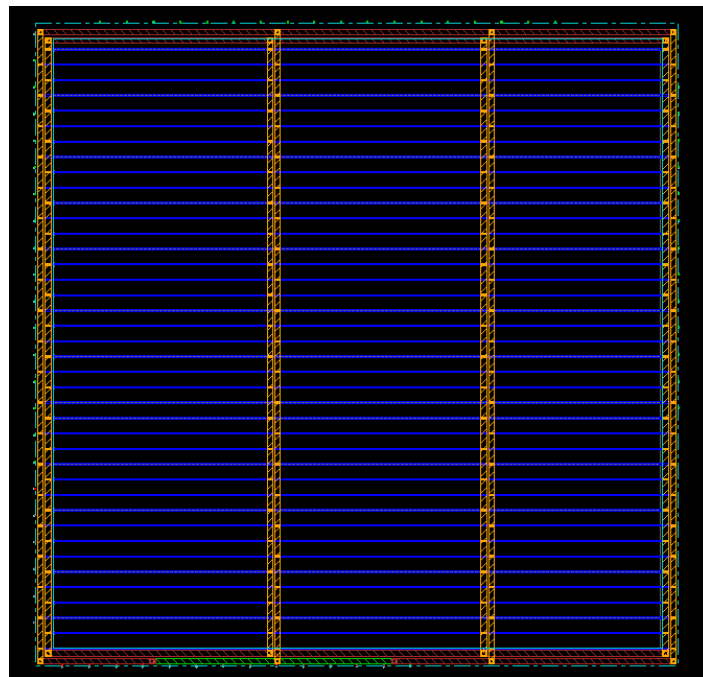
Open pnr folder, open terminal then type encounter

A screenshot of a terminal window titled "Terminal" with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the following commands and output:

```
[localhost.localdomain] ahesham:/home/ahesham/Labs/Lab_PNR_1/pnr$ enc  
enc2xs          enchant          enchant-lsmod encounter  
[localhost.localdomain] ahesham:/home/ahesham/Labs/Lab_PNR_1/pnr$ encounter
```

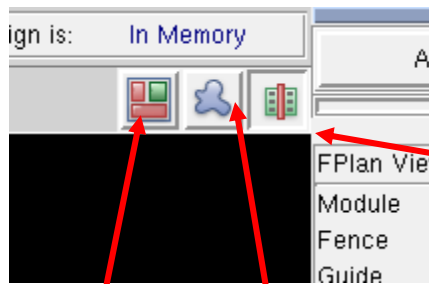
مش لازم تكملها كلها اكتب اول جزء بس ودوس تاب وهي هتكمّل لوحدها

source des_import.tcl and then floorplan.tcl



الحمد لله اهو عملنا لودينج للفلور بلان ورجعناه تاني

Different Views:



Physical view:

شغالين عليه اغلب الوقت

Floor plan view:

لو عايز تشتغل مانيوالي

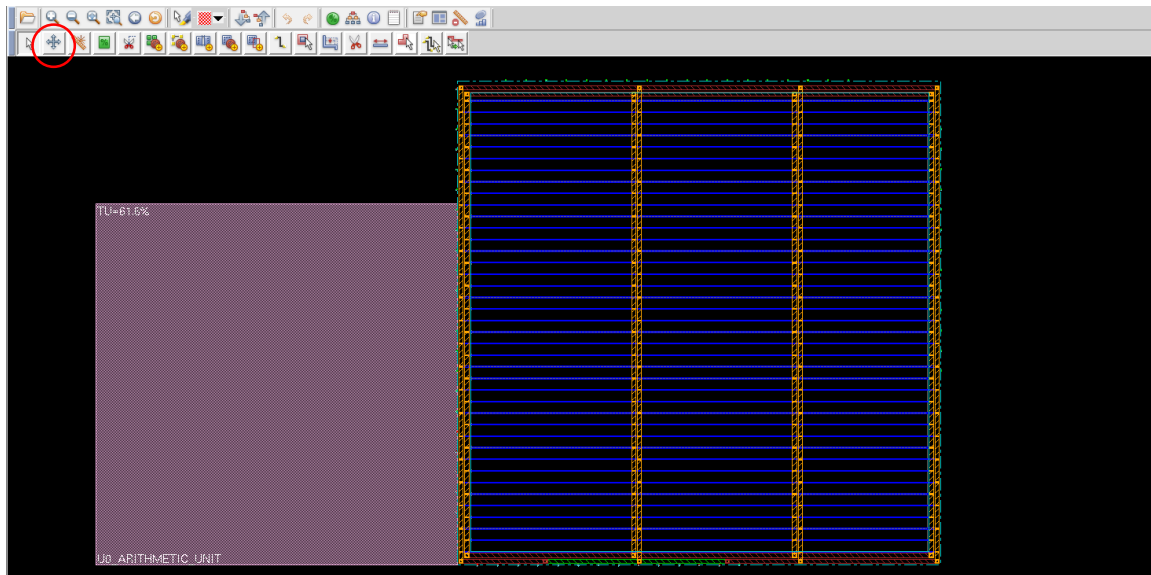
Amoeba view:

بعد الفلورلان البلوكات اتحتطت ازاى

ممکن اچي من التول بار واختار موف

Or click R

واحط البلوكات بنفسي بتحطي بس عادة مش بنشتغل مانيوالي وينسيب التول تتعامل

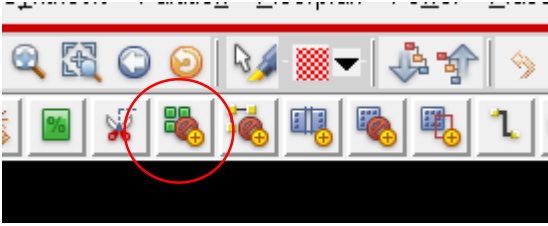


كدة عايزين نرن بليسمنت ودة السكريببت بس قبلها عايزين نعمل حاجة

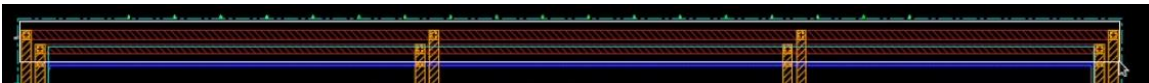
```
##### Placement Standard Cells - TIE High/Low -- connect VDD/VSS ####
placeDesign -inPlaceOpt -prePlaceOpt
addTieHiLo -cell TIELOM -prefix LTIE
addTieHiLo -cell TIEHIM -prefix HTIE
globalNetConnect VDD -type pgpin -pin VDD -inst *
globalNetConnect VSS -type pgpin -pin VSS -inst *
```

Add Placement Blockage

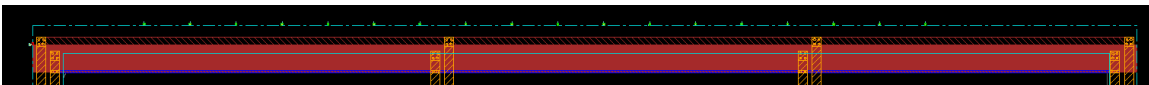
عشان نمنع التول انها تحط ستاندرد سيلز في الحتت اللي مش هيبقا واصل بيه باور وجراوند من فوق وتحت



اختارها وحدد حدود الباوندريز ودوس انتر هتلاقى حط البلوكيدج او حط حطة وكمل الباقي بالاستريتش مود



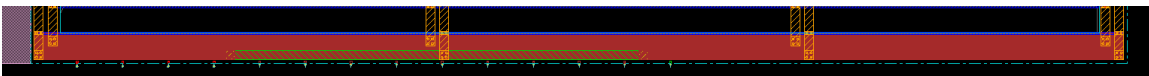
Click Enter



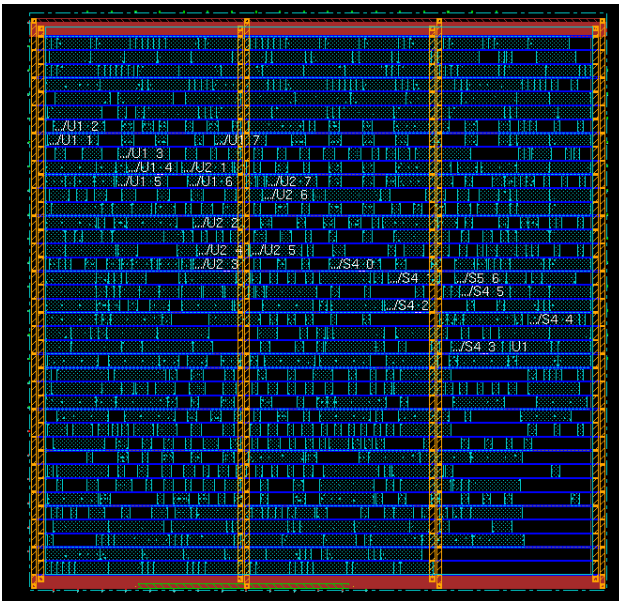
طب دي حاجة هتصنع؟

لا دي حاجة بس عشان اقول للتول متحطيش سيلز هنا مش اكرر

Repeat this step to the bottom also

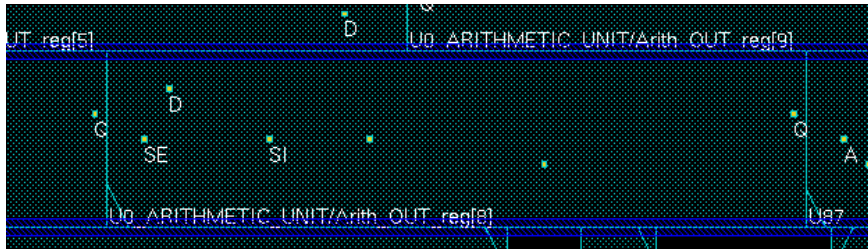


Source placement.tcl



You get this, make sure you are in physical view

It refines placement through many iterations

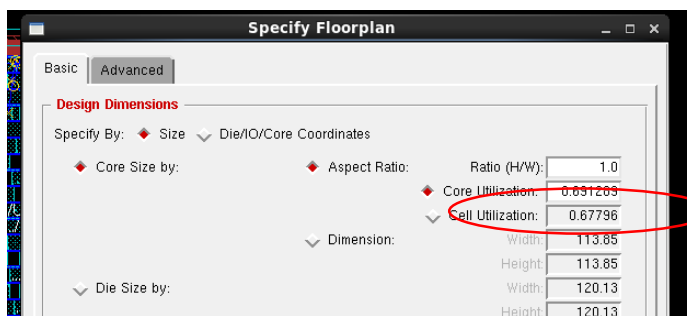
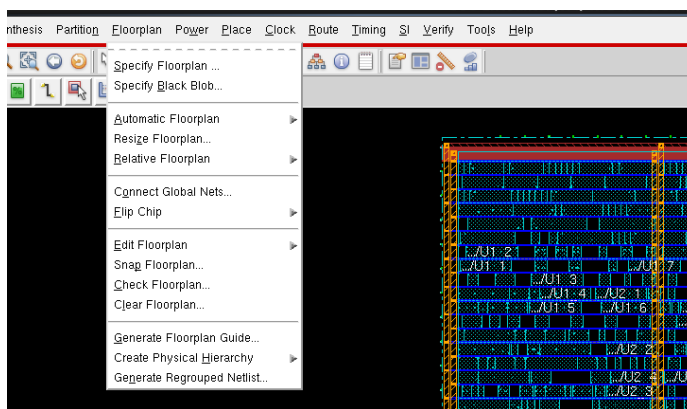


لو خدت بالك من السكان فليب فلوب هتلاقي ليه بينز كثير وكل واحد في حتى فھيھا رخم في البليسمنت

We want to check utilization:

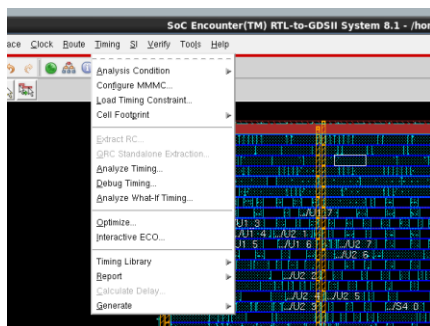
(60 – 75%) in this stage would be ok

Choose floorplan then specify floorplan

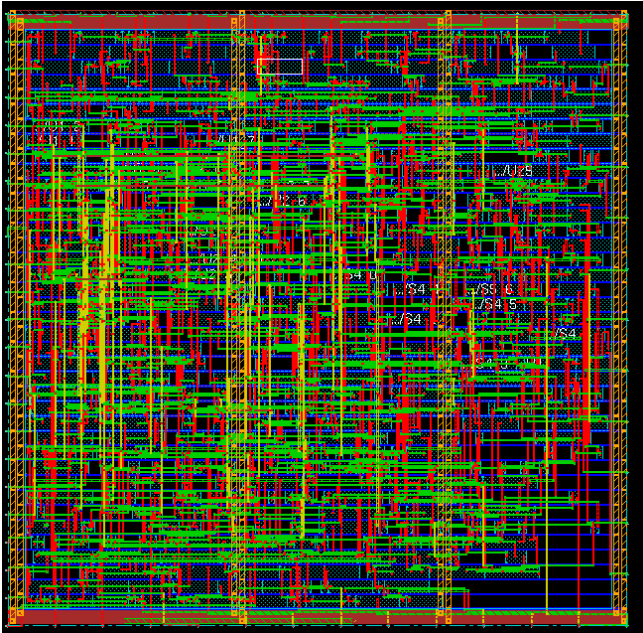


After CTS it would increase as well as upcoming stages so we need to check it after every stage

We want now to run setup analysis



It has no meaning to run setup before placement as no std cells yet



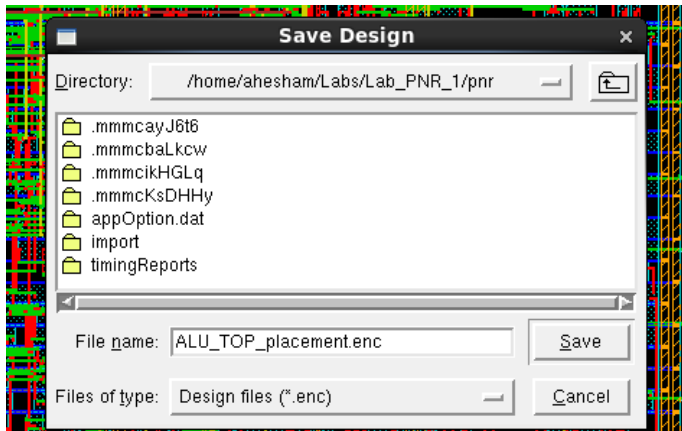
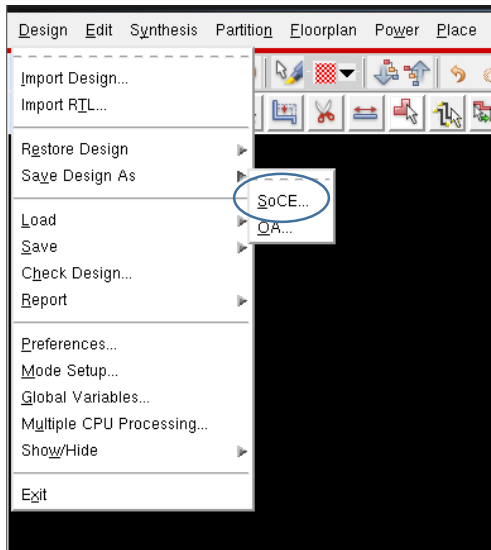
You get this after running

هو بيعمل سيتب اناليسيز علي التوصيل والوايرز اللي التول هتوصلها انما مفيش راوتنج حقيقي لسة

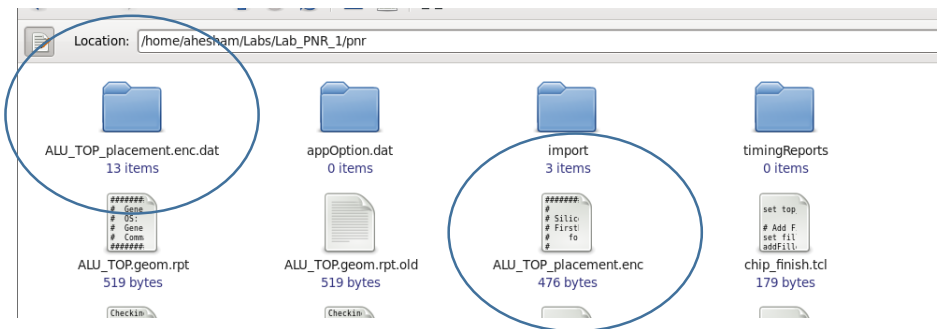
```
-----
timeDesign Summary
-----
+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 47.134 | N/A | 63.894 | 47.134 | N/A | N/A |
| TNS (ns): | 0.000 | N/A | 0.000 | 0.000 | N/A | N/A |
| Violating Paths: | 0 | N/A | 0 | 0 | N/A | N/A |
| All Paths: | 117 | N/A | 78 | 39 | N/A | N/A |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| DRVs | Real | Total | |
|---|---|---|---|
| |Nr nets(terms)| Worst Vio |Nr nets(terms)|
+-----+-----+-----+-----+-----+
| max_cap | 40 (40) | -72.049 | 40 (40) |
| max_tran | 40 (40) | -50.846 | 40 (40) |
| max_fanout | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+-----+
Density: 69.129%
Routing Overflow: 0.00% H and 0.46% V
-----
Reported timing to dir timingReports
Total CPU time: 0.65 sec
Total Real time: 1.0 sec
Total Memory Usage: 384.308594 Mbytes
```

دلوقتي عايز اسيف الديزاين عشان لو لاقدر الله عملت حاجة غلط ارجع ثاني

Design -> Save Design as -> SoCE



You get these file and folder

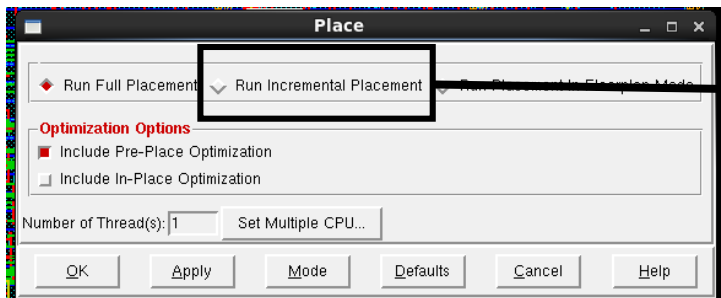
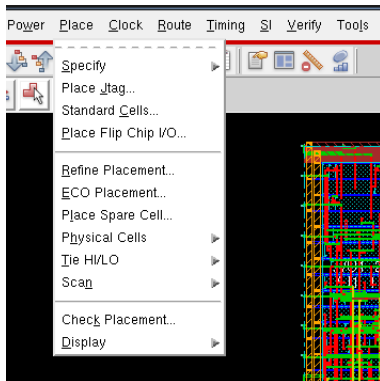


Take away note: We run Hold Analysis after CTS as the skew is not balanced yet

Open timing reports and explore analysis reports

Placement & CTS

Select place then standard cells



Refine Placement

لو هرن ثاني مرة وعايز احسن القديم

بس احنا مش هنشتغل كدة لا هنرن الاسكريبت

CTS Scripts:

```
clockDesign -genSpecOnly Clock.ctstch  
clockDesign -specFile Clock.ctstch -outDir clock_report -fixedInstBeforeCTS
```

Type source cts.tcl or run the 2 commands individually one by one

```
-----  
timeDesign Summary  
-----  
-----  
Setup mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |  
-----  
WNS (ns): | 46.654 | N/A | 64.372 | 46.654 | N/A | N/A |  
TNS (ns): | 0.000 | N/A | 0.000 | 0.000 | N/A | N/A |  
Violating Paths: | 0 | N/A | 0 | 0 | N/A | N/A |  
All Paths: | 117 | N/A | 78 | 39 | N/A | N/A |  
-----  
-----  
DRVs | Real | Total |  
-----  
| Nr nets(terms) | Worst Vio | Nr nets(terms) |  
-----  
max_cap | 40 (40) | -72.049 | 40 (40) |  
max_tran | 40 (40) | -50.846 | 40 (40) |  
max_fanout | 0 (0) | 0 | 0 (0) |  
-----  
Density: 70.230%  
Routing Overflow: 0.00% H and 0.06% V  
-----  
Reported timing to dir clock_report  
Total CPU time: 0.76 sec  
Total Real time: 1.0 sec  
Total Memory Usage: 384.3125 Mbytes  
-----
```

Timing Report

The next and final stage is **Routing**

Run hold analysis post cts then start routhig as follows:

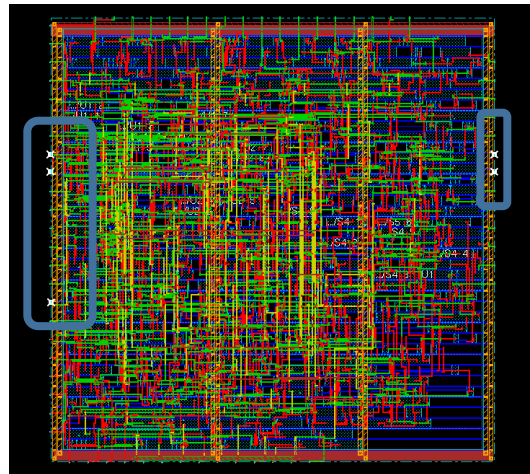
routing.tcl script

```
##### Variables #####
set route_0_optimize_1 0
##### Global/Detailed Routing #####
set max_m_layer 6
if {$route_0_optimize_1 == 0} {
setNanoRouteMode -quiet -routeTopRoutingLayer $max_m_layer
routeDesign -globalDetail -viaOpt -wireOpt
} else {
# Engineering Change Order (ECO) is the process of introducing a minimal localized change to a gate-level netlist or
physical layout to fix a bug, improve performance
refinePlace -preserveRouting
setNanoRouteMode -routeWithEco true
routeDesign -globalDetail -viaOpt -wireOpt
}
```

type: source routing.tcl in terminal

هيقعد يجرب كثير وبخش في اتيريشنز ويحسن سيبه يخلص براحتة

```
#
#Total number of DRC violations = 5
#Total number of net violated process antenna rule = 0
#Total number of violations on LAYER METAL1 = 0
#Total number of violations on LAYER METAL2 = 0
#Total number of violations on LAYER METAL3 = 5
#Total number of violations on LAYER METAL4 = 0
#Total number of violations on LAYER METAL5 = 0
#Total number of violations on LAYER METAL6 = 0
#Total number of violations on LAYER METAL7 = 0
#
#
#detailRoute statistics:
#Cpu time = 00:00:30
#Elapsed time = 00:00:31
#Increased memory = 0.00 (Mb)
#Total memory = 394.00 (Mb)
#Peak memory = 415.00 (Mb)
#Number of warnings = 0
#Total number of warnings = 6
#Number of fails = 0
#Total number of fails = 0
#Complete detailRoute on Thu Sep 18 16:10:01 2025
#
```



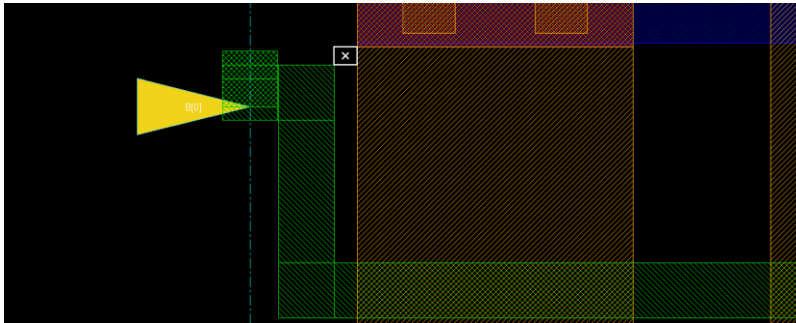
5 Viloations on pins on metal 3

Click X then verify geometry, Do not forget to uncheck Minimum Area option

```
*** Summary ***
Verification Complete : 1 Viols.  0 Wrngs.
*****End: VERIFY GEOMETRY*****
*** verify geometry (CPU: 0:00:00.4 MEM: 5.4M)
```

1 Violation

Real Violation

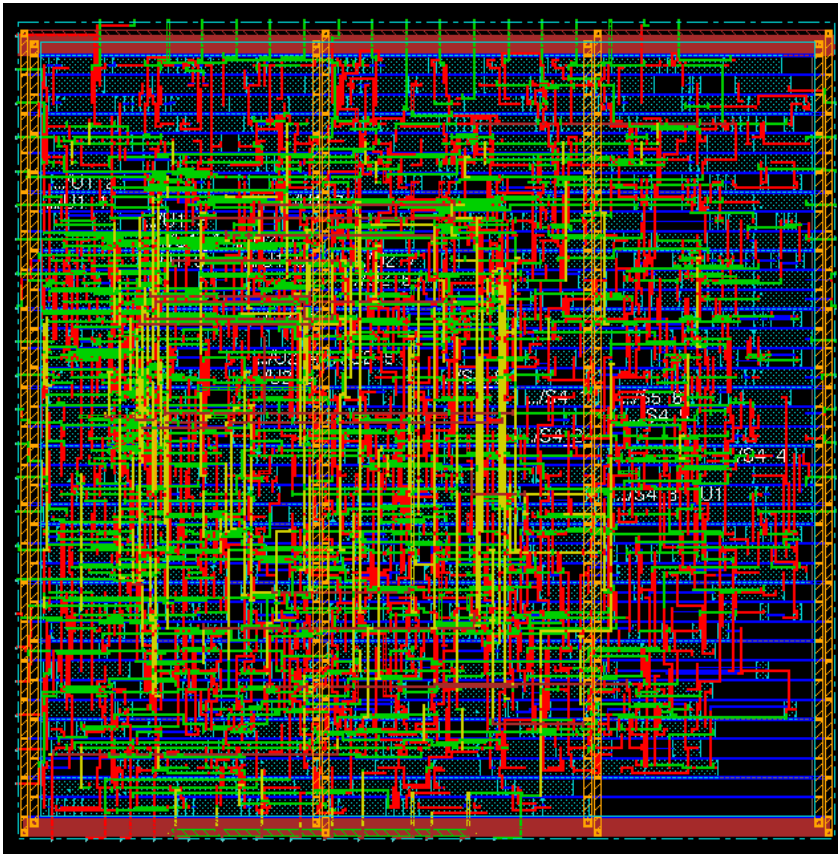


2 Options:

1. Re run routing using using:
set route_0_optimize_1 1
2. Shift the ring a little bit to the right

خلینا مع اول اویشن

Click X and then re run routing



You get this, Zero violations

Run post route setup:

```
-----
timeDesign Summary
-----

+-----+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 47.828 | N/A | 64.588 | 47.828 | N/A | N/A |
| TNS (ns): | 0.000 | N/A | 0.000 | 0.000 | N/A | N/A |
| Violating Paths: | 0 | N/A | 0 | 0 | N/A | N/A |
| All Paths: | 117 | N/A | 78 | 39 | N/A | N/A |
+-----+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+
| DRVs | Real | Total |
| Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+
| max_cap | 40 (40) | -72.048 | 40 (40) |
| max_tran | 40 (40) | -50.845 | 40 (40) |
| max_fanout | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 70.230%
-----
Reported timing to dir timingReports
Total CPU time: 0.58 sec
Total Real time: 1.0 sec
Total Memory Usage: 394.773438 Mbytes
```

Run post route Hold:

```
-----
timeDesign Summary
-----

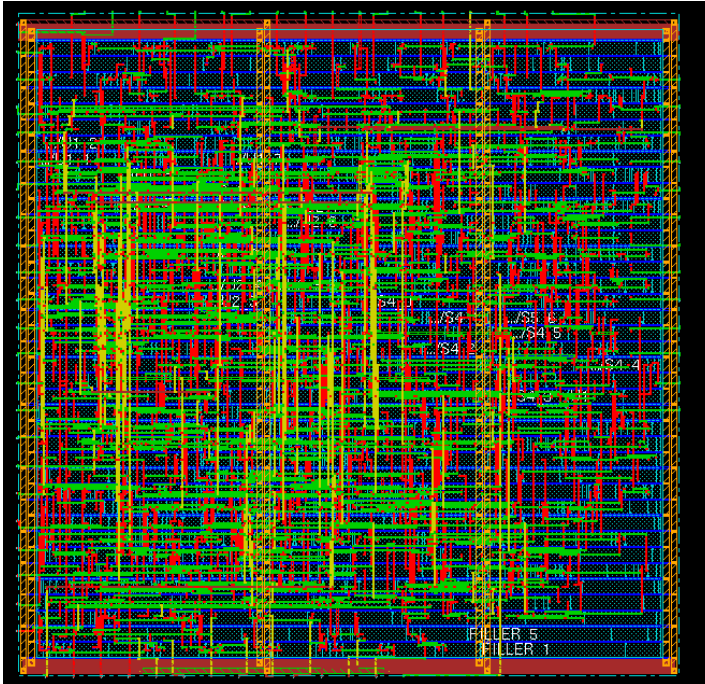
+-----+-----+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 0.271 | N/A | 0.271 | 31.265 | N/A | N/A |
| TNS (ns): | 0.000 | N/A | 0.000 | 0.000 | N/A | N/A |
| Violating Paths: | 0 | N/A | 0 | 0 | N/A | N/A |
| All Paths: | 117 | N/A | 78 | 39 | N/A | N/A |
+-----+-----+-----+-----+-----+-----+-----+

Density: 70.230%
Routing Overflow: 0.00% H and 0.22% V
-----
Reported timing to dir timingReports
Total CPU time: 0.64 sec
Total Real time: 1.0 sec
Total Memory Usage: 394.773438 Mbytes
```

Final Step, Chip finish to add fillers:

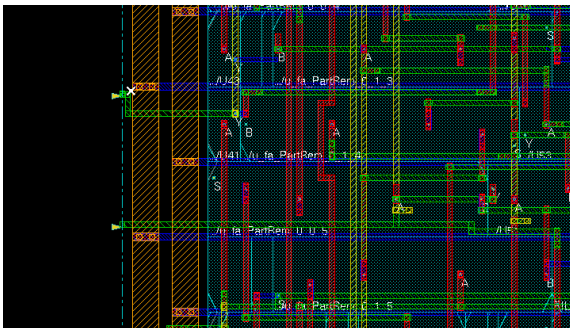
Type source chip_finish.tcl

```
encounter 8> source chip_finish.tcl
**WARN: (SOCSP-9057): Fillers being added in a FIXED mode to the whole design. Some subsequent
operations, such as antenna fixing, may fail due to fillers being marked
as 'FIXED'. If this was not intended, use deleteFiller command to undo.
*INFO: Added 2 filler insts (cell FILL64M / prefix FILLER).
*INFO: Added 9 filler insts (cell FILL32M / prefix FILLER).
*INFO: Added 26 filler insts (cell FILL16M / prefix FILLER).
*INFO: Added 114 filler insts (cell FILL8M / prefix FILLER).
*INFO: Added 210 filler insts (cell FILL4M / prefix FILLER).
*INFO: Added 197 filler insts (cell FILL2M / prefix FILLER).
*INFO: Added 238 filler insts (cell FILL1M / prefix FILLER).
*INFO: Total 796 filler insts added - prefix FILLER (CPU: 0:00:00.0).
*INFO: Checking for DRC violations on added fillers.
*INFO: Iteration 0-#1, Found 692 DRC violations (CPU: 0:00:00.2).
*INFO: Found no DRC violations to fix.
*INFO: Added 0 filler inst (cell FILL1M / prefix FILLER).
*INFO: Iteration 1-#1, Found 692 DRC violations (CPU: 0:00:00.1).
*INFO: End DRC Checks. (CPU: 0:00:00.3 MEM: 0.3M).
```

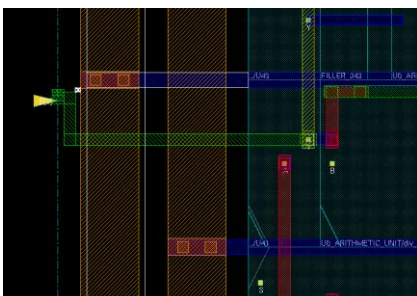


Verify Geometry and Connectivity:

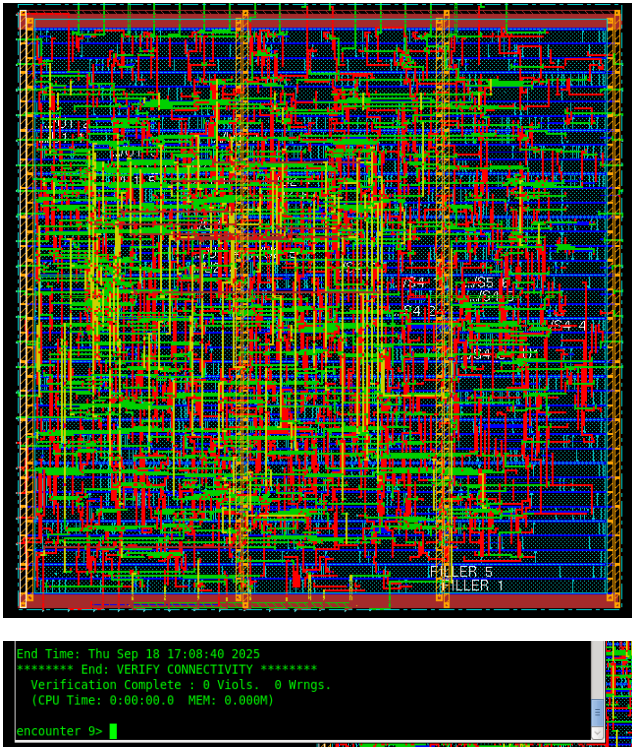
There is the same old violation



Move the power rail to the right a little bit using move wire (m) to avoid silly mistakes, moves horizontally only with all vias

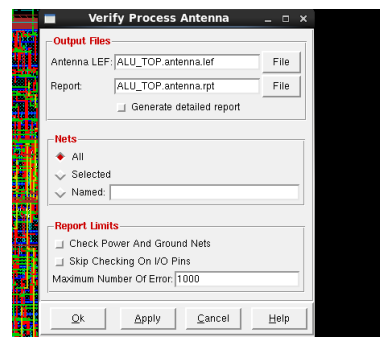
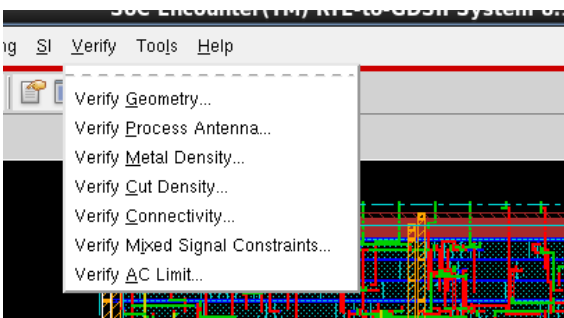


Verify geometry & connectivity



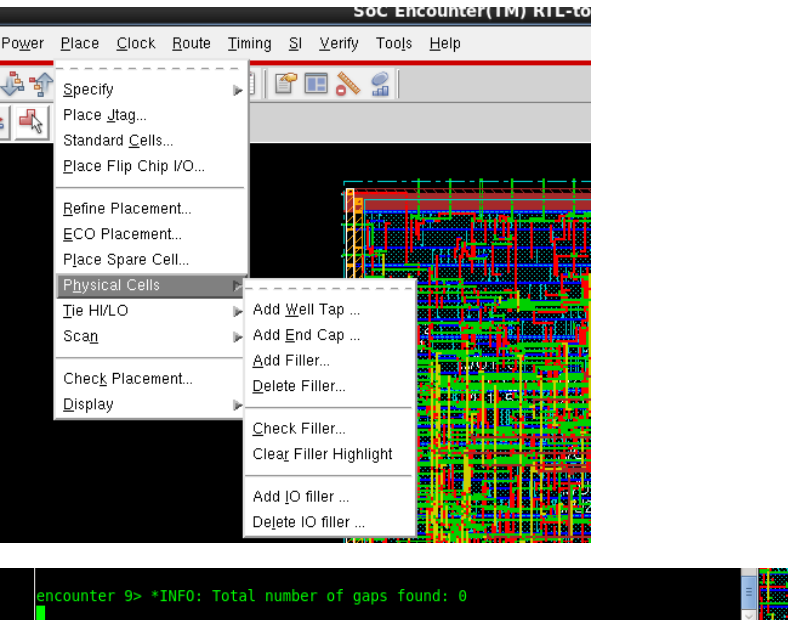
Should be clean no violations

Verify Process Antenna



```
***** START VERIFY ANTENNA *****
Report File: ALU_TOP.antenna.rpt
LEF Macro File: ALU_TOP.antenna.lef
Verification Complete: 0 Violations
***** DONE VERIFY ANTENNA *****
(CPU Time: 0:00:00.1 MEM: 0.000M)
```

Check Fillers:



Last check, Analyze timing setup and hold post-route

```

timeDesign Summary
-----
+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 47.826 | N/A | 64.583 | 47.826 | N/A | N/A |
| TNS (ns): | 0.000 | N/A | 0.000 | 0.000 | N/A | N/A |
| Violating Paths: | 0 | N/A | 0 | 0 | N/A | N/A |
| All Paths: | 117 | N/A | 78 | 39 | N/A | N/A |
+-----+-----+-----+-----+-----+-----+
| DRVs | Real | Total |
| Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+-----+
| max_cap | 40 (40) | -72.848 | 40 (40) |
| max_tran | 40 (40) | -50.845 | 40 (40) |
| max_fanout | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+-----+
Density: 100.000%
Reported timing to dir timingReports
Total CPU time: 0.6 sec
Total Real time: 0.0 sec
Total Memory Usage: 394.398438 Mbytes

```

```

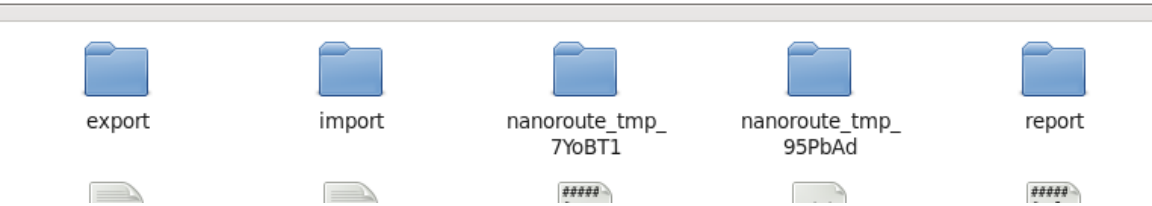
timeDesign Summary
-----
+-----+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 0.018 | N/A | 0.018 | 31.166 | N/A | N/A |
| TNS (ns): | 0.000 | N/A | 0.000 | 0.000 | N/A | N/A |
| Violating Paths: | 0 | N/A | 0 | 0 | N/A | N/A |
| All Paths: | 117 | N/A | 78 | 39 | N/A | N/A |
+-----+-----+-----+-----+-----+-----+
Density: 100.000%
Reported timing to dir timingReports
Total CPU time: 0.63 sec
Total Real time: 1.0 sec
Total Memory Usage: 394.398438 Mbytes

```

Hold report

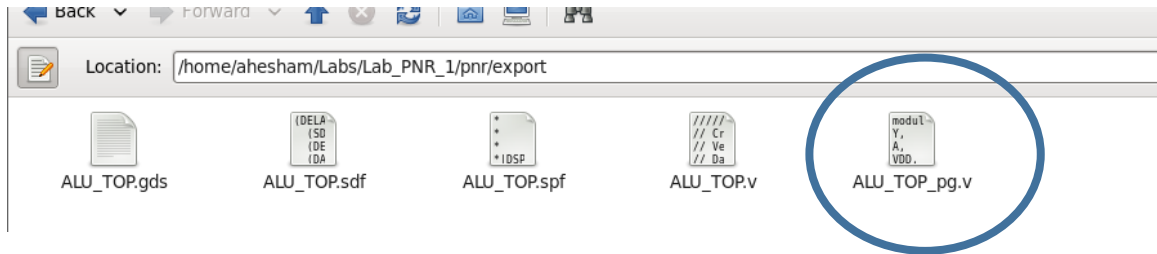
Setup report

Create 2 folders, Export and report:

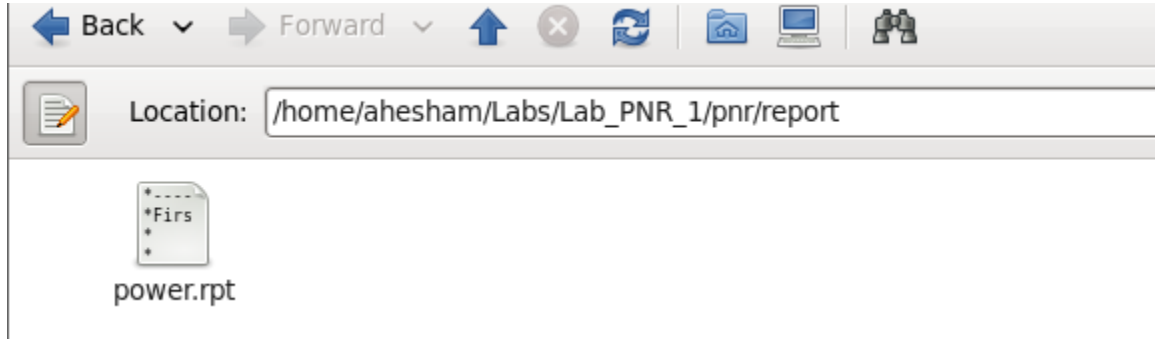


Last Step Type source output_gen.tcl

We get gds, sdf, spf, netlist and netlist with power ground(PG: Contains VDD and VSS)



Final report is power report at report folder:



وبس كدة خلصنا الحمد لله

متساش تدعي لآخواتنا في فلسطين والسودان وكل بقاع الارض ربنا
يخفف عنهم الامهم واحزانهم وينهي معاناتهم وينصرهم علي اعدائهم